

University Of Engineering and Technology, Lahore
Computer Engineering Department

Course Name: Programming Fundamental	Course Code: CMPE-112L
Assignment Type: Lab Report	Dated:
Semester: 1st	Session: 2026 Spring
Lab/Project/Assignment #: 1	CLOs to be covered: CLO
Lab Title: Input/Output, Data type, Operator, If-Else	Teacher Name: Hafiz Muhammad Mubashar

Lab Evaluation:

CLO1	Apply appropriate programming techniques to create executable programs to solve well defined problems					
	Level1	Level2	Level3	Level4	Level5	Level6
(5)						
Total						/5

Rubrics for Current Lab Evaluation

Scale	Marks	Level	Rubric
Excellent	5	L1	Submitted all lab tasks, BONUS task, have good understanding.
Very Good	4	L2	Submitted the lab tasks but have good understanding
Good	3	L3	Submitted the lab tasks but have weak understanding.
Basic	2	L4	Submitted the lab tasks but have no understanding.
Barely Acceptable	1	L5	Submitted only one lab task.
Not Acceptable	0	L6	Did not attempt

LAB No 1

Lab Goals/Objectives:

To understand the basics of Python programming.

Equipment Required:

- Computer/Laptop
- Python 3.x
- IDE (IDLE, VS Code, or PyCharm)

Theory:

Python is a simple, high-level programming language. It uses `input()` for user input and `print()` for output. Python supports data types such as `int`, `float`, `str`, and `bool`. Operators perform arithmetic, comparison, logical, and assignment operations. Conditional statements (`if`, `if-else`, and `if-elif-else`) are used to make decisions based on conditions.

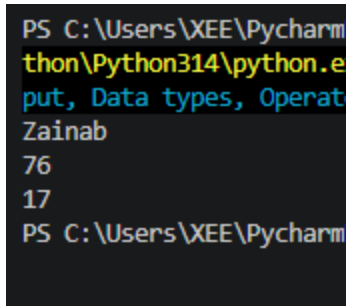
Lab Work:

Input/Output:

Task 1:

```
print("Zainab")
print("76")
print("17")
```

Result:

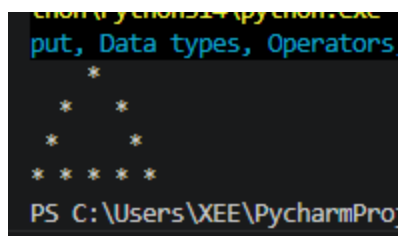


```
PS C:\Users\XEE\Pycharm
thon\Python314\python.exe
put, Data types, Operat
Zainab
76
17
PS C:\Users\XEE\Pycharm
```

Task 2:

```
print("    * ")
print(" * * ")
print(" * * ")
print("* * * * ")
```

Result:



```
thon\Python314\python.exe
put, Data types, Operators
    * 
 * * 
 * * 
* * * * 
PS C:\Users\XEE\PycharmPro
```

Task 3:

```
celsius = float(input("Enter temperature in Celsius: "))
fahrenheit = (celsius * 9/5) + 32
print(celsius, "=", fahrenheit)
```

Result:

```
thon\Python314\python.exe "c:/Users/XEE/PycharmProjects/P
put, Data types, Operators, Conditional Statements (IfEls
Enter temperature in Celsius: 12.4
12.4 = 54.32
PS C:\Users\XEE\PycharmProjects\PythonProject\renewwww>
```

Task 4:

```
Name = input("enter your name")
Age = input("enter your age")
Roll = input(" enter your roll no")
print(Name)
print(Age)
print(Roll)
```

Result:

```
put, Data types, Operators, Conditional State
enter your name Zainab
enter your age 18
enter your roll no 76
Zainab
18
76
PS C:\Users\XEE\PycharmProjects\PythonProject
```

Task 5:

```
bytes = float(input("Enter the number of bytes: "))
megabytes = bytes / (1024 * 1024)
gigabytes = bytes / (1024 * 1024 * 1024)
print("Megabytes (MB):", megabytes)
print("Gigabytes (GB):", gigabytes)
```

Result:

```
put, Data types, Operators, Conditional Statements (IfE
Enter the number of bytes: 23
Megabytes (MB): 2.193450927734375e-05
Gigabytes (GB): 2.1420419216156006e-08
PS C:\Users\XEE\PycharmProjects\PythonProject\renewwww>
```

Task 6:

```
ecat_marks = float(input("Enter your ECAT percentage: "))
inter_marks = float(input("Enter your Intermediate Part-I percentage: "))
matric_marks = float(input("Enter your Matric percentage: "))
aggregate = (ecat_marks * 0.33) + (inter_marks * 0.50) + (matric_marks * 0.17)
print("Your Aggregate is:", round(aggregate, 2), "%")
```

Result:

```
Enter your ECAT percentage: 87
Enter your Intermediate Part-I percentage: 89
Enter your Matric percentage: 92
Your Aggregate is: 88.85 %
PS C:\Users\XEE\PycharmProjects\PythonProject\renewwww>
```

Data Types:

Task 1:

```
age = 20
print(age)
print(type(age))
```

Result:

```
put, Data types, Operato
20
<class 'int'>
PS C:\Users\XEE\Pycharm
```

Task 2:

```
Price = 12.4
print(Price)
print(type(Price))
```

Result:

```
put, Data types, Operators, Cor
12.4
<class 'float'>
PS C:\Users\XEE\PycharmProjects
```

Task 3:

```
name = "zainab"  
print(name)  
print(type(name))
```

Result:

```
zainab  
<class 'str'>  
PS C:\Users\XEE\PycharmProjects\PythonP
```

Task 4:

```
marks = True  
print(marks)  
print(type(marks))
```

Result:

```
put, Data types, Operators, Condit  
True  
<class 'bool'>  
PS C:\Users\XEE\PycharmProjects\Py
```

Operators:

Task 1:

```
a=(int("1"))  
b=(int("2"))  
c=a+b  
print(c)
```

Result:

```
tion\Python314\python.exe - C:\Users\X  
put, Data types, Operators, Condition  
3  
PS C:\Users\XEE\PycharmProjects\Pytho
```

Task 2:

```
a=("1")
b=("2")
c=a+b
print(c)
```

Result:

```
thon\Python314\python.exe "c:/Users/XEE/PycharmProjects/PythonProject/Task2.py"
put, Data types, Operators,
12
PS C:\Users\XEE\PycharmProj
```

Task 3:

```
celsius = float(input("Enter temperature in Celsius: "))
fahrenheit = (celsius * 9/5) + 32
print(celsius, "=", fahrenheit)
```

Result:

```
thon\Python314\python.exe "c:/Users/XEE/PycharmProjects/PythonProject/Task3.py"
put, Data types, Operators, Conditional Statements (IfElse)/3
Enter temperature in Celsius: 24.5
24.5 = 76.1
PS C:\Users\XEE\PycharmProjects\PythonProject\renewwww>
```

Task 4:

```
a=int("5")
b=int("6")
print(a+b)

print(int(3.1))
print(float(3))
```

Result:

```
thon\Python314\python.exe "c:/Users/XEE/PycharmProjects/PythonProject/Task4.py"
put, Data types, Operators,
11
3
3.0
PS C:\Users\XEE\PycharmProje
```

Conditional statement:

Task 1:

```
a=int(input("enter obtained marks"))
b=int(input("enter total marks"))
c=a/b*100
print(c)
if c>=90:
    print("A")
elif c>=85:
    print("A-")
elif c>=80:
    print("B+")
elif c>=75:
    print("B-")
elif c>=70:
    print("C+")
elif c>=65:
    print("C-")
elif c>=60:
    print("D+")
elif c>=55:
    print("D-")
elif c<=50:
    print("F")
```

Result:

```
put, Data types, Operators, Conditional Statements
k 1.py"
enter obtained marks 76
enter total marks 100
76.0
B-
PS C:\Users\XEE\PycharmProjects\PythonProject\rene
```

Task 2:

```
import math
import cmath # Used for imaginary roots

a = float(input("Enter value of a: "))
b = float(input("Enter value of b: "))
c = float(input("Enter value of c: "))

D = b**2 - 4*a*c

print("Discriminant =", D)
if D == 0:
    print("Roots are real, equal and rational.")

    root1 = (-b + math.sqrt(D)) / (2*a)
    root2 = (-b - math.sqrt(D)) / (2*a)

elif D > 0:
    print("Roots are real, distinct and irrational.")

    root1 = (-b + math.sqrt(D)) / (2*a)
    root2 = (-b - math.sqrt(D)) / (2*a)

else:
    print("Roots are imaginary.")

    root1 = (-b + cmath.sqrt(D)) / (2*a)
    root2 = (-b - cmath.sqrt(D)) / (2*a)

print("Root 1 =", root1)
print("Root 2 =", root2)
```

Result:

```
Enter value of a: 3
Enter value of b: 4
Enter value of c: 7
Discriminant = -68.0
Roots are imaginary.
Root 1 = (-0.6666666666666666+1.3743685418725535j)
Root 2 = (-0.6666666666666666-1.3743685418725535j)
PS C:\Users\XEE\PycharmProjects\PythonProject\renewwww>
```

Task 3:

```
num = input("Enter a number: ")

if num == num[::-1]:
    print(num, "is a Palindrome Number")
else:
    print(num, "is Not a Palindrome Number")
```

Result:

```
put, data types, operators, conditional statements (if,
k 3.py"
Enter a number: 6
6 is a Palindrome Number
PS C:\Users\XEE\PycharmProjects\PythonProject\renewwww
```

Task 4:

```
import math

x1 = float(input("Enter x1: "))
y1 = float(input("Enter y1: "))

x2 = float(input("Enter x2: "))
y2 = float(input("Enter y2: "))

def find_quadrant(x, y):
    if x > 0 and y > 0:
        return "First Quadrant"
    elif x < 0 and y > 0:
        return "Second Quadrant"
    elif x < 0 and y < 0:
        return "Third Quadrant"
    elif x > 0 and y < 0:
        return "Fourth Quadrant"
    elif x == 0 and y == 0:
        return "Origin"
    elif x == 0:
        return "On Y-Axis"
    else:
        return "On X-Axis"

q1 = find_quadrant(x1, y1)
q2 = find_quadrant(x2, y2)

distance = math.sqrt((x2 - x1)**2 + (y2 - y1)**2)
print("Point 1 lies in:", q1)
print("Point 2 lies in:", q2)
print("Distance between points =", round(distance, 2))
```

Result:

```
Enter x1: 5
Enter y1: 6
Enter x2: 5
Enter y2: 6
Point 1 lies in: First Quadrant
Point 2 lies in: First Quadrant
Distance between points = 0.0
PS C:\Users\XEE\PycharmProjects\PythonProject\rene
```

Conclusion:

In this lab, we learned the basics of Python programming, including input/output, data types, operators, and conditional statements. These concepts provide the foundation for writing simple and logical Python programs.

University Of Engineering and Technology, Lahore
Computer Engineering Department

Course Name: Programming Fundamentals	Course Code: CMPE-112L
Assignment Type: Lab Report	Dated:
Semester: 1st	Session: 2026 Spring
Lab/Project/Assignment #: 1	CLOs to be covered: CLO
Lab Title: For Loop	Teacher Name: Hafiz Muhammad Mubashar

Lab Evaluation:

CLO1	Apply appropriate programming techniques to create executable programs to solve well defined problems					
	Level1	Level2	Level3	Level4	Level5	Level6
(5)						
Total						/5

Rubrics for Current Lab Evaluation

Scale	Marks	Level	Rubric
Excellent	5	L1	Submitted all lab tasks, BONUS task, have good understanding.
Very Good	4	L2	Submitted the lab tasks but have good understanding
Good	3	L3	Submitted the lab tasks but have weak understanding.
Basic	2	L4	Submitted the lab tasks but have no understanding.
Barely Acceptable	1	L5	Submitted only one lab task.
Not Acceptable	0	L6	Did not attempt

LAB No 1

Lab Goals/Objectives:

- Understand the concept of `for` loops.
- Learn to repeat tasks using loops.
- Use `range()` with `for` loops.

Equipment Required:

- Computer/Laptop
- Python 3.x
- IDE (IDLE, VS Code, or PyCharm)

Theory:

A `for` loop is used to execute a block of code repeatedly for a fixed number of times or to iterate over a sequence such as a list, string, or range. It makes programs shorter, more efficient, and easier to read.

Lab Work:

For Loop:

Task 1:

```
binary = input("Enter a binary number: ")
decimal = 0
power = 0
for digit in reversed(binary):
    decimal += int(digit) * (2 ** power)
    power += 1

print("Decimal Equivalent =", decimal)
```

Result:

```
7 Task 1.py
Enter a binary number: 67
Decimal Equivalent = 19
PS C:\Users\XEE\PycharmProjects\PythonProject\renewwww>
```

Task 2

```
sentence = input("Enter a sentence: ")

vowels = 0
consonants = 0

for ch in sentence:
    ch = ch.lower()

    if ch.isalpha():
        if ch in "aeiou":
            vowels += 1
        else:
            consonants += 1

print("Number of Vowels =", vowels)
print("Number of Consonants =", consonants)
```

Result:

```
/task 2.py"
Enter a sentence: i love pakistan
Number of Vowels = 6
Number of Consonants = 7
PS C:\Users\XEE\PycharmProjects\PythonProject\renewwww>
```

Task 3:

```
start = int(input("Enter starting number: "))
end = int(input("Enter ending number: "))

prime_sum = 0

print("Prime numbers in the given range are:")

for num in range(start, end + 1):
    if num > 1:
        is_prime = True

        for i in range(2, num):
            if num % i == 0:
                is_prime = False
                break

        if is_prime:
            print(num, end=" ")
            prime_sum += num

print("\nSum of prime numbers =", prime_sum)
```

Result:

```
/task 3.py"
Enter starting number: 7
Enter ending number: 87
Prime numbers in the given range are:
7 11 13 17 19 23 29 31 37 41 43 47 53 59 61 67 71 73 79 83
Sum of prime numbers = 864
PS C:\Users\XEE\PycharmProjects\PythonProject\renewwww>
```

Task 4:

```
import random

uppercase = "ABCDEFGHIJKLMNOPQRSTUVWXYZ"
lowercase = "abcdefghijklmnopqrstuvwxyz"
digits = "0123456789"
special = "!@#$%^&*()_+-=[]{}|;:,.<>?/"

length = int(input("Enter password length: "))

use_upper = input("Include uppercase letters? (y/n): ")
use_lower = input("Include lowercase letters? (y/n): ")
use_digits = input("Include digits? (y/n): ")
use_special = input("Include special characters? (y/n): ")

characters = ""

if use_upper.lower() == "y":
    characters += uppercase

if use_lower.lower() == "y":
    characters += lowercase

if use_digits.lower() == "y":
    characters += digits

if use_special.lower() == "y":
    characters += special

if characters == "":
    print("Error: You must select at least one character type!")
else:
    password = ""

    for i in range(length):
        password += random.choice(characters)

    print("Generated Password:", password)
```

Result:

```
/task 4.py"
Enter password length: 8
Include uppercase letters? (y/n): y
Include lowercase letters? (y/n): y
Include digits? (y/n): n
Include special characters? (y/n): n
Generated Password: XJyyuMpc
PS C:\Users\XEE\PycharmProjects\PythonProject\renewwww>
```

Task 5:

```
n = 4
for i in range(1, n + 1):
    for j in range(i):
        print("*", end=" ")
    print()

for i in range(n - 1, 0, -1):
    for j in range(i):
        print("*", end=" ")
    print()
```

Result:

```
/task 5.py"
*
* *
* * *
* * * *
* * *
* *
*
PS C:\Users\XEE\PycharmProjects\Py
```

Task 6:

```
x = int(input("Enter total number of students: "))
for i in range(x):
    print("Student", i + 1)
    name = input("Enter your name: ")
    obtained_marks = int(input("Enter obtained marks: "))
    total_marks = int(input("Enter total marks: "))
    percentage = (obtained_marks / total_marks) * 100
    print("Percentage:", percentage)
    if percentage >= 90:
        grade = "A"
    elif percentage >= 85:
        grade = "A-"
    elif percentage >= 80:
        grade = "B+"
    elif percentage >= 75:
        grade = "B"
    else:
        grade = "C"

    print("Grade:", grade)
```

Result:

```
Task 6.py
Enter total number of students: 2
Student 1
Enter your name: Zainab
Enter obtained marks: 89
Enter total marks: 100
Percentage: 89.0
Grade: A-
Student 2
Enter your name: 
```

Conclusions:

In this lab, we learned how to use `for` loops to repeat tasks efficiently. We also understood how the `range()` function works and how loops simplify repetitive programming tasks.

University Of Engineering and Technology, Lahore
Computer Engineering Department

Course Name: Programming Fundamentals	Course Code: CMPE-112L
Assignment Type: Lab Report	Dated:
Semester: 1st	Session: 2026 Spring
Lab/Project/Assignment #: 1	CLOs to be covered: CLO
Lab Title: While Loop	Teacher Name: Hafiz Muhammad Mubashar

Lab Evaluation:

CLO1	Apply appropriate programming techniques to create executable programs to solve well defined problems					
	Level1	Level2	Level3	Level4	Level5	Level6
(5)						
Total						/5

Rubrics for Current Lab Evaluation

Scale	Marks	Level	Rubric
Excellent	5	L1	Submitted all lab tasks, BONUS task, have good understanding.
Very Good	4	L2	Submitted the lab tasks but have good understanding
Good	3	L3	Submitted the lab tasks but have weak understanding.
Basic	2	L4	Submitted the lab tasks but have no understanding.
Barely Acceptable	1	L5	Submitted only one lab task.
Not Acceptable	0	L6	Did not attempt

LAB No 1

Lab Goals/Objectives:

- Understand the concept of `while` loops.
- Learn to repeat tasks using loops.

Equipment Required:

- Computer/Laptop
- Python 3.x
- IDE (IDLE, VS Code, or PyCharm)

Theory:

A `while` loop repeatedly executes a block of code as long as a given condition is `True`. It is useful when the number of iterations is not known in advance.

Lab Work:

While Loop:

Task 1:

```
i = 1
while i <= 5:
    print(i)
    i += 1
```

Result:

```
op/task 1.py"
1
2
3
4
5
PS C:\Users\XEE\PycharmProjec
```

Task 2

```
total = 0
i = 1
while i <= 10:
    total += i
    i += 1
print("Sum =", total)
```

Result:

```
PS C:\Users\XEE\PycharmProjects\PythonProject\renewwww> & C:\Users\XEE\PycharmProjects\PythonProject\renewwww\Python314\python.exe "c:/Users/XEE/PycharmProjects/PythonProject/op/task 2.py"
Sum = 55
PS C:\Users\XEE\PycharmProjects\PythonProject\renewwww>
```

Task 3:

```
y=int(input("enter number of students"))
i=1
while (i<=y):
    n=input("enter your name")
    a=int(input("enter obtained marks"))
    b=int(input("enter total marks"))
    c=a/b*100
    print(c)
    if c>=90:
        print("A")
    elif c>=85:
        print("A-")
    elif c>=80:
        print("B+")
    elif c>=75:
        print("B-")
    elif c>=70:
        print("C+")
    elif c>=65:
        print("C-")
    elif c>=60:
        print("D+")
    elif c>=55:
        print("D-")
    else :
        print("F")
    i=i+1
```

Result:

```
python3 op/task 3.py
enter number of students 3
enter your name zainab
enter obtained marks 90
enter total marks 100
90.0
A
enter your name
```

Task 4:

```
y=int(input("enter number of students"))
i=0
while (i<=50):
    n=input("enter your name")
    a=int(input("enter obtained marks"))
    b=int(input("enter total marks"))
    c=a/b*100
    print(c)
    if c>=90:
        print("A")
    elif c>=85:
        print("A-")
    elif c>=80:
        print("B+")
    elif c>=75:
        print("B-")
    elif c>=70:
        print("C+")
    elif c>=65:
        print("C-")
    elif c>=60:
        print("D+")
    elif c>=55:
        print("D-")
    else :
        print("F")
    i=i+1
```

Result:

```
python314\python.exe c:/Users/.../
op/task 4.py"
enter number of students 40
enter your name zainab
enter obtained marks 100
enter total marks 120
83.33333333333334
B+
enter your name
```

Conclusions:

In this lab, we learned how to use `while` loops to perform repetitive tasks based on a condition. We also understood how proper loop conditions help control program execution and prevent infinite loops.

University Of Engineering and Technology, Lahore
Computer Engineering Department

Course Name: Programming Fundamentals	Course Code: CMPE-112L
Assignment Type: Lab Report	Dated:
Semester: 1st	Session: 2026 Spring
Lab/Project/Assignment #: 4	CLOs to be covered: CLO
Lab Title: Nested Loop	Teacher Name: Hafiz Muhammad Mubashar

Lab Evaluation:

CLO1	Apply appropriate programming techniques to create executable programs to solve well defined problems					
	Level1	Level2	Level3	Level4	Level5	Level6
(5)						
Total						/5

Rubrics for Current Lab Evaluation

Scale	Marks	Level	Rubric
Excellent	5	L1	Submitted all lab tasks, BONUS task, have good understanding.
Very Good	4	L2	Submitted the lab tasks but have good understanding
Good	3	L3	Submitted the lab tasks but have weak understanding.
Basic	2	L4	Submitted the lab tasks but have no understanding.
Barely Acceptable	1	L5	Submitted only one lab task.
Not Acceptable	0	L6	Did not attempt

LAB No 1

Lab Goals/Objectives:

- Understand the concept of **Nested** loops.
- Learn to repeat tasks using loops.

Equipment Required:

- Computer/Laptop
- Python 3.x
- IDE (IDLE, VS Code, or PyCharm)

Theory:

A nested loop is a loop placed inside another loop. The inner loop executes completely for each iteration of the outer loop. Nested loops are commonly used for creating patterns, working with tables, and processing two-dimensional data.

Lab Work:

Nested Loop:

Task 1:

```
for i in range (1,5):  
    for j in range (3):  
        print(i,j)
```

Result:

```
thon\Python314\p  
oop/task 1.py"  
1 0  
1 1  
1 2  
2 0  
2 1  
2 2  
3 0  
3 1  
3 2  
4 0  
4 1  
4 2  
PS C:\Users\XEEV
```

Task 2

```
for i in range (1,5):  
    for j in range (1):  
        print(i,j)
```

Result:

```
oop/task 2.py"  
1 0  
2 0  
3 0  
4 0  
PS C:\Users\XEEV
```

Task 3:

```
for i in range (1,5):  
    for j in range (i):  
        print("*",end="")  
    print()
```

Result:

```
PS C:\Users\X...  
thon\Python31...  
oop/task 3.py  
*  
**  
***  
****  
PS C:\Users\X...
```

Task 4:

```
for i in range (1,5):  
    for j in range (i):  
        print("i=",i,"j=",j)  
    print(".....")
```

Result:

```
oop/task 4.py"  
i= 1 j= 0  
.....  
i= 2 j= 0  
i= 2 j= 1  
.....  
i= 3 j= 0  
i= 3 j= 1  
i= 3 j= 2  
.....  
i= 4 j= 0  
i= 4 j= 1  
i= 4 j= 2  
i= 4 j= 3  
.....  
PS C:\Users\XEE\PycharmProje
```

Task 5:

```
for i in range (1,5):
    for j in range (i):
        print("(",i,",",j,")",end="")
    print()
```

Result:

```
thon\Python314\pyt
oop/task 5.py"
( 1 , 0 )
( 2 , 0 )( 2 , 1 )
( 3 , 0 )( 3 , 1 )( 3 , 2 )
( 4 , 0 )( 4 , 1 )( 4 , 2 )( 4 , 3 )
PS C:\Users\XEE\PycharmProjects\PythonProject\r
```

Task 6:

```
for i in range (1,5):
    for j in range (1,1+i):
        print(i,end=" ")
    print()
```

Result:

```
thon\Python314\pyt
oop/task 6.py"
1
2 2
3 3 3
4 4 4 4
PS C:\Users\XEE\Py
```

Task 7:

```
for i in range (1,5):
    for j in range (5-i):
        print("*",end=" ")
    print()
```

Result:

```
PS C:\Users\XEE\PycharmProjects\Python314\python.exe C:\Users\XEE\PycharmProjects\Python314\python\Python314\python\oop/task 7.py"
* * * *
* * *
* *
*
PS C:\Users\XEE\PycharmProjects\Python314\python\Python314\python\oop/task 7.py"
```

Task 8:

```
for i in range (1,5):
    for j in range (5-i):
        print("1",end=" ")
    for j in range(i):
        print("*",end=" ")
    print()
```

Result:

```
Python314\python.exe C:\Users\XEE\PycharmProjects\Python314\python\Python314\python\oop/task 8.py"
1 1 1 1 *
1 1 1 * *
1 1 * * *
1 * * * *
1 * * * *
PS C:\Users\XEE\PycharmProjects\Python314\python\Python314\python\oop/task 8.py"
```

Task 9:

```
for i in range (1,5):
    for j in range (5-i):
        print(" ",end=" ")
    for j in range(i*2-1):
        print("*",end=" ")
    print()
```

Result:

```
Python314\python.exe C:\Users\XEE\PycharmProjects\Python314\python\Python314\python\oop/task 9.py"
      *
    * * *
  * * * * *
* * * * * *
PS C:\Users\XEE\PycharmProjects\Python314\python\Python314\python\oop/task 9.py"
```

Task 10:

```
i=1
while i <= 5:
    j=0
    while(j<=5-i):
        print(" ", end=" ")
        j=j+1
    while (j<((i*2)-1)):
        print("*", end=" ")
        j=j+1
    print()
    i=i+1
```

Result:

```
  * *
 * * * *
* * * * *
* * * * *
```

PS C:\Users\XEE\PycharmProjects\Python

Task 11:

```
i=1
while i <= 4:
    j=0
    while(j<=4-i):
        print(" ", end=" ")
        j=j+1
    j=0
    while j<((2*i)-1):
        print("*", end=" ")
        j=j+1
    print( )
    i=i+1
```

Result:

```
python\python314\python.exe c:/Users/
oop/task_11.py"
*
* * *
* * * * *
* * * * * * *
PS C:\Users\XEE\PycharmProjects\
```

Conclusions:

In this lab, we learned how nested loops work and how they can be used to solve complex repetitive tasks. We also practiced creating patterns and improving programming logic using nested loops.

University Of Engineering and Technology, Lahore
Computer Engineering Department

Course Name: Programming Fundamentals	Course Code: CMPE-112L
Assignment Type: Lab Report	Dated:
Semester: 1st	Session: 2026 Spring
Lab/Project/Assignment #: 5	CLOs to be covered: CLO
Lab Title: Functions	Teacher Name: Hafiz Muhammad Mubashar

Lab Evaluation:

CLO1	Apply appropriate programming techniques to create executable programs to solve well defined problems					
	Level1	Level2	Level3	Level4	Level5	Level6
(5)						
Total						/5

Rubrics for Current Lab Evaluation

Scale	Marks	Level	Rubric
Excellent	5	L1	Submitted all lab tasks, BONUS task, have good understanding.
Very Good	4	L2	Submitted the lab tasks but have good understanding
Good	3	L3	Submitted the lab tasks but have weak understanding.
Basic	2	L4	Submitted the lab tasks but have no understanding.
Barely Acceptable	1	L5	Submitted only one lab task.
Not Acceptable	0	L6	Did not attempt

LAB No 1

Lab Goals/Objectives:

- Understand the concept of functions.
- Learn to define and call functions.

Equipment Required:

- Computer/Laptop
- Python 3.x
- IDE (IDLE, VS Code, or PyCharm)

Theory:

A function is a reusable block of code that performs a specific task. In Python, functions are defined using the `def` keyword. Functions can accept parameters, return values, and help make programs organized, readable, and easier to maintain.

Lab Work:

Functions:

Task 1:

```
def factorial(x):
    fact = 1
    for i in range(1, x + 1):
        fact *= i
    return fact

def permutation(n, r):
    return factorial(n) // factorial(n - r)

def combination(n, r):
    return factorial(n) // (factorial(r) * factorial(n - r))

n = int(input("Enter value of n: "))
r = int(input("Enter value of r: "))

print("Permutation (nPr) =", permutation(n, r))
print("Combination (nCr) =", combination(n, r))
```

Result:

```
thon\Python314\python.exe "c:/Users/XEE/PycharmProjects/Pytho
s/task 1.py"
Enter value of n: 6
Enter value of r: 7
Permutation (nPr) = 720
Combination (nCr) = 0
PS C:\Users\XEE\PycharmProjects\PythonProject\renewwww>
```

Task 2

```
large_number = lambda a, b: a if a > b else b

def print_table(num, limit):
    for i in range(1, limit + 1):
        print(f"{num} x {i} = {num * i}")
num1 = int(input("Enter first number: "))
num2 = int(input("Enter second number: "))
limit = int(input("Enter table range: "))
big = large_number(num1, num2)

print("\nLarger number is:", big)
print("Table of larger number:\n")

print_table(big, limit)
```

Result:

```
s/task 2.py
Enter first number: 3
Enter second number: 5
Enter table range: 5

Larger number is: 5
Table of larger number:

5 x 1 = 5
5 x 2 = 10
5 x 3 = 15
5 x 4 = 20
5 x 5 = 25
PS C:\Users\XEE\PycharmProjects\PythonProject\renewwww>
```

Task 3:

```
to_upper = lambda s: s.upper()

def invert(text):
    return text[::-1]

text = input("Enter a string: ")

upper_text = to_upper(text)

print("Uppercase String:", upper_text)

reversed_text = invert(upper_text)

print("Reversed String:", reversed_text)
```

Result:

```
PS C:\Users\XEE\PycharmProjects\PythonProject\renewwww> & C:\Python314\python.exe "c:/Users/XEE/PycharmProjects/PythonProject/task 3.py"
Enter a string: azb
Uppercase String: AZB
Reversed String: BZA
PS C:\Users\XEE\PycharmProjects\PythonProject\renewwww> |
```

Task 4:

```
def celsius_to_fahrenheit(c):
    return (c * 9/5) + 32

def fahrenheit_to_celsius(f):
    return (f - 32) * 5/9

print("Temperature Conversion Program")
print("1. Celsius to Fahrenheit")
print("2. Fahrenheit to Celsius")

choice = int(input("Enter your choice (1 or 2): "))

if choice == 1:
    c = float(input("Enter temperature in Celsius: "))
    print("Fahrenheit =", celsius_to_fahrenheit(c))

elif choice == 2:
    f = float(input("Enter temperature in Fahrenheit: "))
    print("Celsius =", fahrenheit_to_celsius(f))

else:
    print("Invalid choice!")
```

Result:

```
C:\Python314\python.exe "c:/Users/XEE/PycharmProjects/PythonProject/task 4.py"
Temperature Conversion Program
1. Celsius to Fahrenheit
2. Fahrenheit to Celsius
Enter your choice (1 or 2): 2
Enter temperature in Fahrenheit: 23.9
Celsius = -4.500000000000001
PS C:\Users\XEE\PycharmProjects\PythonProject\renewwww> |
```

Task 5:

```
def calculate_gpa(courses):
    total_points = 0
    total_credits = 0

    for gp, ch in courses:
        total_points += gp * ch
        total_credits += ch

    if total_credits == 0:
        return 0

    return total_points / total_credits

n = int(input("Enter number of courses: "))

courses = []

for i in range(n):
    gp = float(input(f"Enter grade point for course {i+1}: "))
    ch = float(input(f"Enter credit hours for course {i+1}: "))
    courses.append((gp, ch))
gpa = calculate_gpa(courses)

print("Your GPA =", round(gpa, 2))
```

Result:

```
PS C:\Users\XEE\PycharmProjects\PythonProject\renewwww> & C:\Users\XEE\PycharmProjects\PythonProject\renewwww\Python314\python.exe "c:/Users/XEE/PycharmProjects/PythonProject/renewwww/s/task 5.py"
Enter number of courses: 2
Enter grade point for course 1: 12
Enter credit hours for course 1: 3
Enter grade point for course 2: 12
Enter credit hours for course 2: 2
Your GPA = 12.0
PS C:\Users\XEE\PycharmProjects\PythonProject\renewwww> █
```

Task 6:

```
def calculate_gpa():
    total_grade_points = 0
    total_credit_hours = 0

    n = int(input("Enter number of courses: "))

    for i in range(n):
        gp = float(input(f"Enter grade point for course {i+1}: "))
        ch = float(input(f"Enter credit hours for course {i+1}: "))

        total_grade_points += gp * ch
        total_credit_hours += ch

    if total_credit_hours == 0:
        return 0

    return total_grade_points / total_credit_hours

gpa = calculate_gpa()

print("Your GPA =", round(gpa, 2))
```

Result:

```
PS C:\Users\XEE\PycharmProjects\PythonProject\renewwww> & C:\Users\XEE\AppData\Local\Microsoft\WindowsApps\PythonSoftwareFoundation\Python314\python.exe "c:/Users/XEE/PycharmProjects/PythonProject/task 6.py"
Enter number of courses: 2
Enter grade point for course 1: 10
Enter credit hours for course 1: 2
Enter grade point for course 2: 13
Enter credit hours for course 2: 1
Your GPA = 11.0
PS C:\Users\XEE\PycharmProjects\PythonProject\renewwww> |
```

Task 7:

```
def n(name):
    print(f"hello {name} , welcome to python")
n("ali")
n("sara")
n("ahmed")
```

Result:

```
PS C:\Users\XEE\PycharmProjects\PythonProject\renewwww> & C:\Us
thon\Python314\python.exe "c:/Users/XEE/PycharmProjects/PythonPr
s/task 7.py"
hello ali , welcome to python
hello sara , welcome to python
hello ahmed , welcome to python
PS C:\Users\XEE\PycharmProjects\PythonProject\renewwww>
```

Task 8:

```
def sum(a,b):  
    return a+b  
c=1  
d=1  
a=sum(c,d)  
print(a)  
  
b=sum(3,4)  
print(b)
```

Result:

```
thon\Python314\python.exe "c:/Users/XEE/Py
5/task 8.py"
2
7
PS C:\Users\XEE\PycharmProjects\PythonPro
```

Task 9:

[illegible]

Result:

```
hello world
hello world
hello world
hello world
hello world
hello world
hello world
hello world
hello world
hello world
hello world
hello world
hello world
hello world
hello world
hello world
hello world
PS C:\Users\XEE\PycharmProjects\PythonProject\renewwww>
```

Conclusions:

In this lab, we learned how to create and use functions in Python. We also understood the importance of functions in reducing code repetition and improving program structure.

University Of Engineering and Technology, Lahore
Computer Engineering Department

Course Name: Programming Fundamentals	Course Code: CMPE-112L
Assignment Type: Lab Report	Dated:
Semester: 1st	Session: 2026 Spring
Lab/Project/Assignment #: 6	CLOs to be covered: CLO
Lab Title: List, Tuple, Dictionary	Teacher Name: Hafiz Muhammad Mubashar

Lab Evaluation:

CLO1	Apply appropriate programming techniques to create executable programs to solve well defined problems					
	Level1	Level2	Level3	Level4	Level5	Level6
(5)						
Total						/5

Rubrics for Current Lab Evaluation

Scale	Marks	Level	Rubric
Excellent	5	L1	Submitted all lab tasks, BONUS task, have good understanding.
Very Good	4	L2	Submitted the lab tasks but have good understanding
Good	3	L3	Submitted the lab tasks but have weak understanding.
Basic	2	L4	Submitted the lab tasks but have no understanding.
Barely Acceptable	1	L5	Submitted only one lab task.
Not Acceptable	0	L6	Did not attempt

LAB No 1

Lab Goals/Objectives:

- Understand lists, tuples, and dictionaries.
- Learn to store and access data using these data structures.

Equipment Required:

- Computer/Laptop
- Python 3.x
- IDE (IDLE, VS Code, or PyCharm)

Theory:

Lists, tuples, and dictionaries are built-in Python data structures. A **list** is an ordered and mutable collection of items. A **tuple** is an ordered but immutable collection. A **dictionary** stores data as key-value pairs, allowing fast and efficient data retrieval.

Lab Work:

List:

Task 1:

```
l=[1,"ali",45]
l.append(5)
print(l)
```

Result:

```
C:\Python314\python.exe
ple, Dictionary/1. List/task
[1, 'ali', 45, 5]
PS C:\Users\XEE\PycharmProj
```

Task 2

```
fruits = ["apple", "banana", "mango"]
print(fruits)
```

Result:

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

PS C:\Users\XEE\PycharmProjects\PythonProject\renewwww> & C:\Python314\python.exe "c:/Users/XEE/PycharmProjects/PythonProject/Dictionary/1. List/task 2.py"
['apple', 'banana', 'mango']
PS C:\Users\XEE\PycharmProjects\PythonProject\renewwww> |
```

Task 3:

```
fruits = ["apple", "banana", "mango"]
fruits.remove("banana")
print(fruits)
```

Result:

```
PS C:\Users\XEE\PycharmProjects\PythonProject\renewwww> & C:\Users\XEE\PycharmProjects\PythonProject\Python314\python.exe "c:/Users/XEE/PycharmProjects/PythonProject/Dictionary/1. List/task 3.py"
['apple', 'mango']
PS C:\Users\XEE\PycharmProjects\PythonProject\renewwww>
```

Task 4:

```
numbers = [10, 20, 30, 40, 50]
print(max(numbers))
print(min(numbers))
print(sum(numbers))
```

Result:

```
thon\Python314
ple, Dictionary
50
10
150
PS C:\Users\XEE\
```

Tuples:

Task 1:

```
fruits = ("apple", "banana", "mango")
print(fruits)
```

Result:

```
thon\Python314\python.exe "c:/Users/XEE/PycharmProjects/PythonProject/Dictionary/2. Tuple/task 1.py"
('apple', 'banana', 'mango')
PS C:\Users\XEE\PycharmProjects\PythonProject\renewwww>
```

Task 2:

```
my_tuple = (1, 2, 3)
print(type(my_tuple))
```

Result:

```
PS C:\Users\XEE\PycharmProjects\PythonProject\renewwww> & C:\Users\XEE\PycharmProjects\PythonProject\Python314\python.exe "c:/Users/XEE/PycharmProjects/PythonProject/Dictionary/2. Tuple/task 4.py"
<class 'tuple'>
PS C:\Users\XEE\PycharmProjects\PythonProject\renewwww>
```

Dictionary:

Task 1:

```
a={"apple":"saeb","mango":"aam","grapes":"'angoor'"}  
print(a["apple"])
```

Result:

```
Python 3.14.0 (tags/v3.14.0:2024/08/24) [AMD64]  
File, Dictionary/3. Dictionary/task 1.py  
saeb  
PS C:\Users\XEE\PycharmProjects\PythonProject\renewwww>
```

Task 2:

```
a={"apple":"saeb","mango":"aam","grapes":"'angoor'"}  
print(a["apple"])  
print(a["mango"])  
a["dates"]="khajoor"  
print(a)
```

Result:

```
Python 3.14.0 (tags/v3.14.0:2024/08/24) [AMD64]  
File, Dictionary/3. Dictionary/task 2.py  
saeb  
aam  
{'apple': 'saeb', 'mango': 'aam', 'grapes': 'angoor', 'dates': 'khajoor'}  
PS C:\Users\XEE\PycharmProjects\PythonProject\renewwww>
```

Task 3:

```
student = {"name": "Ali", "age": 20}  
student["city"] = "Lahore"  
print(student)
```

Result:

```
Python 3.14.0 (tags/v3.14.0:2024/08/24) [AMD64]  
File, Dictionary/3. Dictionary/task 3.py  
{'name': 'Ali', 'age': 20, 'city': 'Lahore'}  
PS C:\Users\XEE\PycharmProjects\PythonProject\renewwww>
```

Conclusions:

In this lab, we learned how to use lists, tuples, and dictionaries to store and manage data in Python. We also practiced performing basic operations on these data structures and understood their different uses.

University Of Engineering and Technology, Lahore
Computer Engineering Department

Course Name: Programming Fundamentals	Course Code: CMPE-112L
Assignment Type: Lab Report	Dated:
Semester: 1st	Session: 2026 Spring
Lab/Project/Assignment #: 7	CLOs to be covered: CLO
Lab Title: Sets	Teacher Name: Hafiz Muhammad Mubashar

Lab Evaluation:

CLO1	Apply appropriate programming techniques to create executable programs to solve well defined problems					
	Level1	Level2	Level3	Level4	Level5	Level6
(5)						
Total						/5

Rubrics for Current Lab Evaluation

Scale	Marks	Level	Rubric
Excellent	5	L1	Submitted all lab tasks, BONUS task, have good understanding.
Very Good	4	L2	Submitted the lab tasks but have good understanding
Good	3	L3	Submitted the lab tasks but have weak understanding.
Basic	2	L4	Submitted the lab tasks but have no understanding.
Barely Acceptable	1	L5	Submitted only one lab task.
Not Acceptable	0	L6	Did not attempt

LAB No 1

Lab Goals/Objectives:

- Understand the concept of sets.
- Learn to create and use sets in Python

.

Equipment Required:

- Computer/Laptop
- Python 3.x
- IDE (IDLE, VS Code, or PyCharm)

Theory:

A **set** is an unordered collection of unique elements in Python. Sets are used to store distinct values and support operations such as union, intersection, difference, and membership testing. They are useful for removing duplicate values and performing mathematical set operations.

Lab Work:

Set:

Task 1:

```
s={}
a=type(s)
print(a)
print(s)
#{ } is non-empty set it becomes dictionary

b=set()
type(b)
print(b)
# set() is actual empty set
# 0/0 or infinity/infinity is indeterminent form bcz they cant be defined
```

Result:

```
Python 3.14 (python.exe) C:\Users\XEE\PycharmProjects\PythonProject\1.py
1.py
<class 'dict'>
{}
set()
PS C:\Users\XEE\PycharmProjects\PythonProject\renewwww>
```

Task 2

```
s={1,2,3,"zainab"}
s.add("noor")
s.discard("zainab")
s.remove(1)
s.discard("ali")
print(s)
#unlike discard if we write something non-existing by remove it gives error
```

Result:

```
PS C:\Users\XEE\PycharmProjects\PythonProject\renewwww> & C:\Users\XEE\AppData\Local\Microsoft\Windows\Python314\python.exe "c:/Users/XEE/PycharmProjects/PythonProject/renewwww/Week 6/2.py"
{2, 3, 'noor'}
PS C:\Users\XEE\PycharmProjects\PythonProject\renewwww> █
```

Task 3:

```
name={"noor","zainab","tayyaba","ayesha"}
for i in name:
    user = input("enter your name:")
    if user in name:
        print("exist")
    else:
        print("not exist")
```

Result:

```
PS C:\Users\XEE\PycharmProjects\PythonProject\renewwww> & C:\Users\XEE\AppData\Local\Microsoft\Windows\Python314\python.exe "c:/Users/XEE/PycharmProjects/PythonProject/renewwww/Week 6/3.py"
enter your name:Ali
not exist
enter your name:█
```

Task 4:

```
s={1,2,3,"zainab"}
s.pop()
print(s)
s.clear()
print(s)
```

Result:

```
PS C:\Users\XEE\PycharmProjects\PythonProject\renewwww> & C:\Users\XEE\AppData\Local\Microsoft\Windows\Python314\python.exe "c:/Users/XEE/PycharmProjects/PythonProject/renewwww/Week 6/4.py"
{2, 3, 'zainab'}
set()
PS C:\Users\XEE\PycharmProjects\PythonProject\renewwww> █
```

Task 5:

```
s1={1,2,3}
s2={4,5,6}
s3={7,8,9}
b=set.union(s1,s2,s3)
print(b)
```

Result:

```
PS C:\Users\XEE\PycharmProjects\PythonProject\renewwww> & C:\Users\XEE\PycharmProjects\PythonProject\Python314\python.exe "c:/Users/XEE/PycharmProjects/PythonProject/renewwww/5.py"
{1, 2, 3, 4, 5, 6, 7, 8, 9}
PS C:\Users\XEE\PycharmProjects\PythonProject\renewwww> █
```

Task 6:

```
s1={1,2,3}
s2={4,5,6}
s3={7,8,9}
s2.update(s3)
s1.update(s2)

print(s1)
print(s2)
print(s3)
```

Result:

```
PS C:\Users\XEE\PycharmProjects\PythonProject\renewwww> & C:\Users\XEE\PycharmProjects\PythonProject\Python314\python.exe "c:/Users/XEE/PycharmProjects/PythonProject/renewwww/6.py"
{1, 2, 3, 4, 5, 6, 7, 8, 9}
{4, 5, 6, 7, 8, 9}
{8, 9, 7}
PS C:\Users\XEE\PycharmProjects\PythonProject\renewwww> █
```

Task 7:

```
s1={1,2,3}
s2={4,2,6}
s3={7,2,9}
b=set.intersection(s1,s2,s3)
print(b)
c=s1.intersection_update(s1,s2,s3)
print(c)
```

Result:

```
PS C:\Users\XEE\PycharmProjects\PythonProject\renewwww> & C:\Users\XEE\PycharmProjects\PythonProject\Python314\python.exe "c:/Users/XEE/PycharmProjects/PythonProject/renewwww/7.py"
{2}
None
PS C:\Users\XEE\PycharmProjects\PythonProject\renewwww> █
```

Task 8:

```
s1={1,2,3}
s2={4,2,6}
b=set.isdisjoint(s1,s2)
print(b)
```

Result:

```
thon\Python314\python.exe "c:/Users/
8.py"
False
PS C:\Users\XEE\PycharmProjects\Pyt
```

Task 9:

```
s1={1,2,3}
s2={4,2,6}
b=s1.issubset(s2)
print(b)
```

Result:

```
9.py"
False
PS C:\Users\XEE\PycharmProjects\PythonProject\renewwww>
```

Task 10:

```
s1={1,2,3}
s2={4,2,6}
b=set.symmetric_difference(s1,s2)
print(b)
```

Result:

```
PS C:\Users\XEE\PycharmProjects\PythonProject\renewwww> & C:\Users\XEE
thon\Python314\python.exe "c:/Users/XEE/PycharmProjects/PythonProject/r
10.py"
{1, 3, 4, 6}
PS C:\Users\XEE\PycharmProjects\PythonProject\renewwww>
```

Task 11:

```
s={1,2,2,3}
a={"ali","ali"}
print(a)
print(s)
b=set([1,2,3,7])
print(b)
```

Result:

```
thon\Python314\python.exe "c:/Users/XEE/PycharmProjects/Pyth
11.py"
{'ali'}
{1, 2, 3}
{1, 2, 3, 7}
PS C:\Users\XEE\PycharmProjects\PythonProject\renewwww>
```

Conclusions:

In this lab, we learned how to create and use sets in Python. We also practiced basic set operations and understood how sets help manage unique data efficiently.

University Of Engineering and Technology, Lahore
Computer Engineering Department

Course Name: Programming Fundamentals	Course Code: CMPE-112L
Assignment Type: Lab Report	Dated:
Semester: 1st	Session: 2026 Spring
Lab/Project/Assignment #: 8	CLOs to be covered: CLO
Lab Title: Recursion	Teacher Name: Hafiz Muhammad Mubashar

Lab Evaluation:

CLO1	Apply appropriate programming techniques to create executable programs to solve well defined problems					
	Level1	Level2	Level3	Level4	Level5	Level6
(5)						
Total						/5

Rubrics for Current Lab Evaluation

Scale	Marks	Level	Rubric
Excellent	5	L1	Submitted all lab tasks, BONUS task, have good understanding.
Very Good	4	L2	Submitted the lab tasks but have good understanding
Good	3	L3	Submitted the lab tasks but have weak understanding.
Basic	2	L4	Submitted the lab tasks but have no understanding.
Barely Acceptable	1	L5	Submitted only one lab task.
Not Acceptable	0	L6	Did not attempt

LAB No 1

Lab Goals/Objectives:

- Understand the concept of recursion.
- Learn to write recursive functions.

Equipment Required:

- Computer/Laptop
- Python 3.x
- IDE (IDLE, VS Code, or PyCharm)

Theory:

Recursion is a programming technique in which a function calls itself to solve a problem. Every recursive function must have a **base case** to stop the recursion and prevent infinite function calls. It is commonly used for problems that can be divided into smaller subproblems.

Lab Work:

Recursion:

Task 1:

```
def n (a,b):  
    s = a+b  
    return s  
v=n (1,2)  
print(v)
```

Result:

```
PS C:\Users\XEE\PycharmProjects\PythonProject\renewwww> & C:  
thon\Python314\python.exe "c:/Users/XEE/PycharmProjects/Python  
n/task 1.py"  
3  
PS C:\Users\XEE\PycharmProjects\PythonProject\renewwww> █
```

Task 2

```
def show (n):  
    print(n)  
    if (n == 0):  
        return  
  
    show (n - 1)  
  
show(6)
```

Result:

```
thon\Python314\python  
n/task 2.py"  
6  
5  
4  
3  
2  
1  
0  
PS C:\Users\XEE\Pychar
```

Task 3:

```
def show(n):  
    if (n==0) or (n==1):  
        return 1  
    else:  
        return show(n - 1)  
print(show(6))
```

Result:

```
PS C:\Users\XEE\PycharmProjects\PythonProject\renewwww> & C:\Users\XEE\AppData\Local\Microsoft\WindowsApps\Python\Python314\python.exe "c:/Users/XEE/PycharmProjects/PythonProject/task 3.py"  
1  
PS C:\Users\XEE\PycharmProjects\PythonProject\renewwww> |
```

Conclusions:

In this lab, we learned how recursion works by creating recursive functions. We also understood the importance of the base case and how recursion can simplify solving certain programming problems.

University Of Engineering and Technology, Lahore
Computer Engineering Department

Course Name: Programming Fundamentals	Course Code: CMPE-112L
Assignment Type: Lab Report	Dated:
Semester: 1st	Session: 2026 Spring
Lab/Project/Assignment #: 9	CLOs to be covered: CLO
Lab Title: Exception Handling	Teacher Name: Hafiz Muhammad Mubashar

Lab Evaluation:

CLO1	Apply appropriate programming techniques to create executable programs to solve well defined problems					
	Level1	Level2	Level3	Level4	Level5	Level6
(5)						
Total						/5

Rubrics for Current Lab Evaluation

Scale	Marks	Level	Rubric
Excellent	5	L1	Submitted all lab tasks, BONUS task, have good understanding.
Very Good	4	L2	Submitted the lab tasks but have good understanding
Good	3	L3	Submitted the lab tasks but have weak understanding.
Basic	2	L4	Submitted the lab tasks but have no understanding.
Barely Acceptable	1	L5	Submitted only one lab task.
Not Acceptable	0	L6	Did not attempt

LAB No 1

Lab Goals/Objectives:

- Understand the concept of exception handling.
- Learn to use `try`, `except`, and `finally` blocks.

Equipment Required:

- Computer/Laptop
- Python 3.x
- IDE (IDLE, VS Code, or PyCharm)

Theory:

Exception handling is used to manage errors that occur during program execution. Python uses `try`, `except`, `else`, and `finally` blocks to catch and handle exceptions, allowing the program to continue running without crashing.

Lab Work:

Exception Handling:

Task 1:

```
try:
    age = int(input("Enter your age: "))
    print(f"Your age is {age}")
except ValueError:
    print("Please enter a number, not letters!")
```

Result:

```
PS C:\Users\XEE\PycharmProjects\PythonProject\renewwww> & C:\Users\XEE\AppData\Local\Microsoft\WindowsApps\Python\Python314\python.exe "c:/Users/XEE/PycharmProjects/PythonProject/Exception Handling/task 1.py"
Enter your age: 18
Your age is 18
PS C:\Users\XEE\PycharmProjects\PythonProject\renewwww> █
```

Task 2

```
try:
    a=int(input("enter a number"))
    print(a)
except:
    print("invalid")
    print("hello world")
```

Result:

```
PS C:\Users\XEE\PycharmProjects\PythonProject\renewwww> & C:\Users\XEE\AppData\Local\Microsoft\WindowsApps\Python\Python314\python.exe "c:/Users/XEE/PycharmProjects/PythonProject/Exception Handling/task 2.py"
enter a number 4
4
PS C:\Users\XEE\PycharmProjects\PythonProject\renewwww> █
```

Conclusions:

In this lab, we learned how to handle runtime errors using exception handling in Python. We also understood how `try`, `except`, and `finally` blocks make programs more robust and user-friendly.

University Of Engineering and Technology, Lahore
Computer Engineering Department

Course Name: Programming Fundamentals	Course Code: CMPE-112L
Assignment Type: Lab Report	Dated:
Semester: 1st	Session: 2026 Spring
Lab/Project/Assignment #: 10	CLOs to be covered: CLO
Lab Title: Lambda, Os Library, Map	Teacher Name: Hafiz Muhammad Mubashar

Lab Evaluation:

CLO1	Apply appropriate programming techniques to create executable programs to solve well defined problems					
	Level1	Level2	Level3	Level4	Level5	Level6
(5)						
Total						/5

Rubrics for Current Lab Evaluation

Scale	Marks	Level	Rubric
Excellent	5	L1	Submitted all lab tasks, BONUS task, have good understanding.
Very Good	4	L2	Submitted the lab tasks but have good understanding
Good	3	L3	Submitted the lab tasks but have weak understanding.
Basic	2	L4	Submitted the lab tasks but have no understanding.
Barely Acceptable	1	L5	Submitted only one lab task.
Not Acceptable	0	L6	Did not attempt

LAB No 1

Lab Goals/Objectives:

- Understand lambda (anonymous) functions.
- Learn to create and use lambda expressions.

Equipment Required:

- Computer/Laptop
- Python 3.x
- IDE (IDLE, VS Code, or PyCharm)

Theory:

A **lambda function** is a small anonymous function defined using the `lambda` keyword. It is used for simple operations without creating a named function. The **os library** provides functions to interact with the operating system, such as creating, renaming, deleting files and folders, and accessing directory information.

Lab Work:

Lambda Functions:

Task 1:

```
def cube(n):  
    return n**3  
print(cube(5))  
def square(n):  
    return n**2  
print(square(5))
```

Result:

```
PS C:\Users\XEE\PycharmProjects\PythonProject\renewwww> python  
Python\Python314\python.exe "c:/Users/XEE/PycharmProjects/PythonProject/renewwww.py"  
function , module-based program , os modules/1. lambda function  
125  
25  
PS C:\Users\XEE\PycharmProjects\PythonProject\renewwww>
```

Task 2

```
a=lambda x:x*x*x  
print(a(5))
```

Result:

```
function , module-based program , os modules/1. lambda function  
125  
PS C:\Users\XEE\PycharmProjects\PythonProject\renewwww>
```

Task 3

```
def sum(x,y,z):  
    return(x+y+z)  
x=1  
y=2  
z=3  
print(sum(x,y,z))
```

Result:

```
PS C:\Users\XEE\PycharmProjects\PythonProject\renewwww> & C:\Python314\python.exe "c:/Users/XEE/PycharmProjects/PythonProject/lambda_functions/1. lambda function.py"
6
PS C:\Users\XEE\PycharmProjects\PythonProject\renewwww>
```

Task 4

```
a=lambda x,y,z:x+y+z
print(a(5,4,6))
```

Result:

```
thon\Python314\python.exe "c:/Users/XEE/PycharmProjects/PythonProject/lambda_functions/4. lambda functions/task4.py"
15
PS C:\Users\XEE\PycharmProjects\PythonProject\renewwww>
```

Task 5

```
a=[1,2,3]
for i in range(len(a)):
    a[i]=a[i]**2
print(a)
```

Result:

```
thon\Python314\python.exe "c:/Users/XEE/PycharmProjects/PythonProject/lambda_functions/5. lambda functions/task5.py"
[1, 4, 9]
PS C:\Users\XEE\PycharmProjects\PythonProject\renewwww>
```

Task 6

```
f=open("pf.txt","a")
a=f.write("noon")
print(a)
f.close()
```

Result:

```
thon\Python314\python.exe "c:/Users/XEE/PycharmProjects/PythonProject/lambda_functions/6. lambda functions/task6.py"
4
PS C:\Users\XEE\PycharmProjects\PythonProject\renewwww>
```

Import Os libraries:

Text File 1:

r=read

w=write

a=append

b=binary file

Text File 2:

r=read

w=write

a=append

b=binary file

f=open("15.txt")

a=f.read()

print(a)

f.close()

Task 1:

```
import os
for i in range(0,100):
    os.mkdir(f"data{i+1}")
```

Result:

```
> data1
> data2
> data3
> data4
> data5
> data6
> data7
> data8
> data9
> data10
> data11
> data12
> data13
> data14
> data15
> data16
> data17
> data18
> data19
> data20
> data21
> data22
> data23
> data24
> data25
> data26
```

```
1 import os
2 for i in range(0,100):
3     os.mkdir(f"data{i+1}")
```

PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS Python + - [] [X]

```
PS C:\Users\XEE\PycharmProjects\PythonProject\renewwww> & C:\Users\XEE\AppData\Local\Microsoft\Windows\Python314\python.exe "c:/Users/XEE/PycharmProjects/PythonProject/renewwww/Week
unction , module-based program , os modules/1. lambda functions/task 7.py"
4
PS C:\Users\XEE\PycharmProjects\PythonProject\renewwww> & C:\Users\XEE\AppData\Local\Microsoft\Windows\Python314\python.exe "c:/Users/XEE/PycharmProjects/PythonProject/renewwww/Week
unction , module-based program , os modules/2. Module based programs/import os.py"
PS C:\Users\XEE\PycharmProjects\PythonProject\renewwww> 
```

Task 2

```
import zainab
zainab.hello()
```

Result:

```
unction , module-based program , os modules/2. Module based program
Hello im here
PS C:\Users\XEE\PycharmProjects\PythonProject\renewwww> 
```

Task 3

```
def hello():  
    print("Hello im here")  
if __name__ == "__main__" :  
    hello()
```

Result:

```
PS C:\Users\XEE\PycharmProjects\PythonProject\renewwww> & C:\Users\XEE\AppData\Local\Pr  
thon\Python314\python.exe "c:/Users/XEE/PycharmProjects/PythonProject/renewwww/Week 10.  
unction , module-based program , os modules/2. Module based programs/zainab.py"  
Hello im here  
PS C:\Users\XEE\PycharmProjects\PythonProject\renewwww> █
```

Os Module:

Task 1

```
import os  
  
print(os.getcwd())
```

Result:

```
PS C:\Users\XEE\PycharmProjects\PythonProject\renewwww> & C:\Users\XEE  
thon\Python314\python.exe "c:/Users/XEE/PycharmProjects/PythonProject/  
unction , module-based program , os modules/3. os modules examples/tas  
C:\Users\XEE\PycharmProjects\PythonProject\renewwww  
PS C:\Users\XEE\PycharmProjects\PythonProject\renewwww> █
```

Task 2

```
import os  
  
os.mkdir("NewFolder")  
print("Folder created")
```

Result:

```
PS C:\Users\XEE\PycharmProjects\PythonProject\renewwww> & C:\Users\XEE  
thon\Python314\python.exe "c:/Users/XEE/PycharmProjects/Python  
unction , module-based program , os modules/3. os modules exa  
Folder created  
PS C:\Users\XEE\PycharmProjects\PythonProject\renewwww> █
```

Task 3

```
import os

print(os.listdir())
```

Result:

```
ata35', 'data36', 'data37', 'data38', 'data39', 'data4', 'data40', 'data41', 'data42', 'data43',
'data44', 'data45', 'data46', 'data47', 'data48', 'data49', 'data5', 'data50', 'data51', 'data52',
, 'data53', 'data54', 'data55', 'data56', 'data57', 'data58', 'data59', 'data6', 'data60', 'data6
1', 'data62', 'data63', 'data64', 'data65', 'data66', 'data67', 'data68', 'data69', 'data7', 'dat
a70', 'data71', 'data72', 'data73', 'data74', 'data75', 'data76', 'data77', 'data78', 'data79', '
data8', 'data80', 'data81', 'data82', 'data83', 'data84', 'data85', 'data86', 'data87', 'data88',
'data89', 'data9', 'data90', 'data91', 'data92', 'data93', 'data94', 'data95', 'data96', 'data97
', 'data98', 'data99', 'NewFolder', 'pf.txt', 'Week 01. InputOutput, Data types, Operators, Condi
tional Statements (IfElse)', 'Week 02. for Loop', 'Week 03. while loop', 'Week 04. Nested Loop',
'Week 05. Functions', 'Week 06. List, Tuple, Dictionary', 'Week 07. Set', 'Week 08. Recursion', '
Week 09. Exception Handling', 'Week 10. lambda function , module-based program , os modules']
PS C:\Users\XEE\PycharmProjects\PythonProject\renewwww>
```

Conclusions:

In this lab, we learned how to use lambda functions for simple operations and explored the `os` library for performing file and directory management tasks. These concepts help write concise and efficient Python programs.

University Of Engineering and Technology, Lahore
Computer Engineering Department

Course Name: Programming Fundamentals	Course Code: CMPE-112L
Assignment Type: Lab Report	Dated:
Semester: 1st	Session: 2026 Spring
Lab/Project/Assignment #: 10	CLOs to be covered: CLO
Lab Title: QT Designer	Teacher Name: Hafiz Muhammad Mubashar

Lab Evaluation:

CLO1	Apply appropriate programming techniques to create executable programs to solve well defined problems					
	Level1	Level2	Level3	Level4	Level5	Level6
(5)						
Total						/5

Rubrics for Current Lab Evaluation

Scale	Marks	Level	Rubric
Excellent	5	L1	Submitted all lab tasks, BONUS task, have good understanding.
Very Good	4	L2	Submitted the lab tasks but have good understanding
Good	3	L3	Submitted the lab tasks but have weak understanding.
Basic	2	L4	Submitted the lab tasks but have no understanding.
Barely Acceptable	1	L5	Submitted only one lab task.
Not Acceptable	0	L6	Did not attempt

LAB No 1

Lab Goals/Objectives:

- Understand the purpose of Qt Designer.
- Learn to create graphical user interfaces (GUIs).
- Design windows, buttons, labels, and input fields.
- Integrate Qt Designer with Python programs..

Equipment Required:

- Computer/Laptop
- Python 3.x
- Qt Designer
- PyQt5/PySide6
- IDE (VS Code, PyCharm, or IDLE)

Theory:

Qt Designer is a visual tool used to create graphical user interfaces (GUIs) by dragging and dropping widgets. The designed interface is saved as a `.ui` file, which can be loaded into Python using **PyQt5** or **PySide6** to build interactive desktop applications.

Lab Work:

Qt Designer:

Task 1

Login page for University Portal

Result:



Conclusions:

In this lab, we learned how to design user interfaces using Qt Designer and connect them with Python applications. This enables the development of user-friendly and interactive desktop programs.